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Short Note

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Sub Name: Embedded system
Sub code: ECA 4416

1. Worst case interrupt latency in CAN frame Reception.

A cortex-M7 running at 480 MHz has a clock period if:

$$T_{clk} = \frac{1}{480MHz} \simeq 2.08\ ns$$

The 35% branch frequency can cause pipeline stalls because branch instructions may change the program flow and requires the pipeline to be refilled.

So, cache Misses increases interrupt latency. If the required ISR Instructions or data are not present in the instruction / data cache.

Therefore, the worst case latency is Approximately

$$Latency = (N_{base} + N_{branch} + N_{cache}) \times 2.08\ ns$$

2. Root cause and fix

The intermittent is mainly caused by high CPU utilization (97%) after adding image processing task. this leaves only about 3% CPU capacity for interrupts, operating tasks, and other control functions.

As a Result task scheduling delays, interrupt latency and possible watchdog timeout can occur.

The 192 KB RAM may also become a problem if the image-processing task requires large buffers or causes stack/heap exhaustion.

3. RMS schedulability using Liu & Layland Bound

for 3 periodic table. the Liu & Layland utilization bound is,

$$U_{LL} = n(2^{1/n} - 1)$$

$$U_{LL} = 3(2^{1/3} - 1) \approx 0.779$$

So, the task set is guaranteed scheduled under Rate Monotonic scheduling (RMS) if,

$$U \leq 77.9\%$$

The sample rates are:

ECG : 500Hz → Period - 2 ms

SPO_2 : 50 Hz → 20 ms

Temperature : 1Hz → 1 s

$$\therefore U = \sum \frac{C_i}{T_i}$$

If ≤ 0.779 the system is guaranteed scheduled under RMS. if it exceeds 77.9% schedulability is not guaranted and the tasks must be further analysis.

Priority order: ECG > SPO_2 > Temperature

energy Bottle and Duty-cycle Redesign

The main bottleneck is excessive avg Power consumption compared with the available solar energy. A 3000 mAH battery lasting only 18 days gives an average consumption of

$$I_{avg} = \frac{3000}{15 \times 24} = 6.94 \text{ mA}$$

For the winter operation, the solar input is only 20 mA, so the system must keep its average consumption below 20 mA to become energy.

A suitable duty-cycle is to keep the controller and sensors in deep-sleep mode most of the time and wake them periodically for sensing, processing and communication. High Power activities such as WiFi / LoRa transmission.

$I_{avg} < 20$ mA

FSM safety Analysis

undefined transitions in a railway - crossing FSM are a safety hazard because the controller may enter an unpredictable state or fail to activate the warning.

For three undefined transitions:

1. Train detected ⇒ undefined: Hazard of failing to close barrier
2. Barrier closing ⇒ undefined: Hazard of leaving the barier in an unsafe intermediate condition.
3. Train cleared: Hazards of keeping the barrier closed / opened it before the train has safely.

guard: train_cleared == 1 AND Track_clear == 1 ⇒ opening state